MILITARY SPECIFICATION
CASE, CARTRIDGE ASSEMBLY, 40MM, M169, INERT METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of case cartridge designated as Case Cartridge Assembly, 40MM, M169, Inert.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition

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MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9275800 - Case CartridgeAssembly, 40MM, M169, Inert

PUBLICATIONS

ARMY

ET-9275800 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64 - Standard Method of Salt Spray (Fog) Testing

ASTM E 18-67 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

ASTM E 8-69 - Tension Testing of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

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3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.- The parts shall comply with all requirements specified on Drawing (Dwg.) 9275764, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Coating thickness. The thickness of the anodic coating shall be a minimum of .3 mils (.003 inches).

3.4 Hardness.- The hardness of the area near the mouth and on the base of the cartridge case shall be 67 ± 7 on the Rockwell 30 T Scale.

3.5 Workmanship.- All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (see 6.4).

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference

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shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this speification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Fifty (50) cartridge cases (5 for salt spray).
b. Twenty (20) base plugs (5 for salt spray).
c. Six (6) test specimens of cartridge case for tensile properties.
d. Twenty (20) case cartridge assemblies.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials,

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including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment), material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.- All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and the same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105.

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Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When ths method is elected, it shall be applie to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Base plug (See Dwg. 8886332).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None Defined.		
Major:	AQL 1.50 Percent		
101.	Largest outside diameter	Gage	01001
102.	Concentricity of smallest outside diameter with largest outside diameter	Gage	01002
103.	Concentricity of primer cavity diameter with largest outside diameter	Gage	01003
104.	Smallest outside diameter	Gage	01004
105.	Diameter of primer cavity	Gage	01005

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Categories	Defects	Method of Inspection	Code No.
	106. Depth of primer cavity	Gage	01006
	107. Diameter of flash hole	Gage	01007
	108. Length below shoulder	Gage	01008
	109. Radius at intersection of primer cavity missing	Visual	01009
	110. Length from primer end to bottom of shoulder	Gage	01010
	111. Length between shoulders, excluding rear angle	Gage	01011
	112. Location of small datum diameter	Gage	01012
	113. Location of large datum diameter	Gage	01013
Minor:	AQL 2.50 Percent		
	201. Second largest outside diameter	Gage	01014
	202. Protective coating with bare spot except anode rack marks	Visual	01015
	203. Finish improper	Visual	01016
	204. Evidence of poor workmanship (see 3.5)	Visual	01017
	205. Color incorrect	Visual	01018

4.3.2.2 Case Cartridge (See Dwg. 8886329)

Categories	Defects	Method of Inspection	Code No.
Major:	AQL 2.50 Percent		
	101. *Hardness (See Para. 4.4.1)*	Test	02001
	102. Concentricity of flange diameter with second largest outside diameter	Gage	02002
	103. Concentricity of largest inside diameter open end, with second largest outside diameter	Gage	02003
	104. Concentricity of base plug cavity with second largest outside diameter	Gage	02004
	105. Concentricity of propellant charge cavity with base plug cavity	Gage	02005

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Categories	Defects	Method of Inspection	Code No.
Major:			
106.	Diameter of propellant charge cavity, straight portion	Gage	02006
107.	Diameter of base plug cavity	Gage	02007
108.	Depth of base plug cavity	Gage	02008
109.	Second largest outside diameter (See 6.5)	Gage	02009
110.	Perpendicularity of open end to second largest outside diameter	Gage	02010
111.	Perpendicularity of base to second largest outside diameter	Gage	02011
112.	Diameter of flange	Gage	02012
113.	Diameter below flange	Gage	02013
114.	Total length	Gage	02014
115.	Diameter of vent holes	Gage	02015
116.	Largest inside diameter, open end	Gage	02016
117.	Location of large datum diameter, propellant cavity	Gage	02017
118.	Location of small datum diameter, propellant cavity	Gage	02018
119.	Thickness through node	Gage	02019
120.	Thickness of flange	Gage	02020
121.	Depth of sealing lip	Gage	02021
122.	Location of large datum diameter on node	Gage	02022
123.	Finish of vent holes improper	Gage	02023
124.	Burr in vent hole	Visual Equip	02024
125.	One vent hole blocked or missing	Visual	02025
126.	Location of vent holes	Gage	02026
127.	Metal defective (see 6.6 and 3.5)	Visual	02027
128.	More than one vent hole blocked or missing	Visual	02028

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.50 Percent		
201.	Protective coating with bare spot except anode rack marks	Visual	02029
202.	Color incorrect	Visual	02030
203.	Evidence of poor workmanship except burrs (see 3.5)	Visual	02031

4.3.2.3 Case cartridge assembly (See dwg.) 9275764 .

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Base plug, above flush	Gage	03001
102.	Depth to base plug, maximum (max.)	Gage	03002
103.	Holes missing	Visual	03003
Minor:	AQL 1.50 Percent		
201.	RTV not flush	Visual	03004
202.	Incorrect number of holes	Visual	03005
203.	Evidence of poor workmanship (see 3.5)	Visual	03006

4.3.3 Testing.

4.3.3.1 Salt spray (See Dwg. 8886329 and 8886332), Major defect, Code No. 04001,- Sampling and rejection shall be in accordance with Schedule A of MIL-A-8625. The test shall be performed as specified in 4.4.2.

4.3.3.2 Tensile properties of case cartridge (see Dwg. 8886329), Major defect, Code No. 05001.- Sampling and rejection shall be in accordance with ASTM-B-221. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.3 Process control inspection of bath temperature for anodic coating (see Dwgs. 8886329 and 8886332), Major defect, Code No. (06001). - Continuous monitoring inspection shall be conducted for bath temperature requirement specified on the applicable drawings. If the bath temperature for a batch of parts does not comply with the applicable drawing requirement, the batch shall be suitably identified classified defective and removed from the lot.

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4.3.3.4 Anodic coating thickness (See Para. 3.3 , Major defect, Code No. 07001.- The sample size for batch processes shall be selected in accordance with MIL-STD-105. Inspection level II, using an AQL of .40 Percent. The sample size for a continuous process shall be selected in accordance with MIL-STD-1235, Inspection level II, using an AQL of .40 Percent. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment. - Equipment Tabulation (ET number 9275764 and Equipment Lists (EL's referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Hardness.- The case cartridge shall be subjected to the Rockwell hardness test in accordance with ASTM-E-18. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Salt spray.- The case cartridge and base plug shall be subjected to the salt spray test in accordance with ASTM-B-117 using 5 percent salt solution. This test shall be conducted for 240 hours. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Tensile properties of cartridge case. - The shape of the specimen for determining cartridge case tensile strength shall be as specified on the Equipment List. The specimen shall be taken prior to trimming and anodizing from a 3/4 inch wide hoop, 1/4 inch in from the mouth of the cartridge case. The hoop shall be split, unrolled and cut to the required shape. The test

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shall be conducted in accordance with ASTM-E-8-69. Tensile strength shall be computed as indicated in above mentioned method. Yield strength will be computed using a 2% offset. Each specimen must exhibit a minimum tensile strength of 50,000 psi and a minimum yield strength of 45,000 psi.

4.4.4 Anodic coating thickness. - The equipment used to measure the thickness shall be as specified on the Equipment List and in accordance with the operating and calibration instructions provided. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.- The case cartridge assembly, inert shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level C.- The case cartridge assembly, inert, preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping.- When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

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6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment Lists.- Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010 the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU" and non-mandatory (indicated by drawing or Federal Stock Number . When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4 or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to

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completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contract reproductions. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.- All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, Attn: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests. - Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SMUPA-QA-A-S.

6.5 If the requirement cannot be met in the free state, the "go" inside diameter plug gage may be inserted in the cartridge case while inspecting the "go" and "no go" second largest outside diameter. If the "go" inside diameter plug gage is utilized, the data cards shall be annotated accordingly.

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6.6 Metal defective.- Nick, crack, dent, or sharp edge which might interfere with the proper assembly or end item performance of the item.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A016

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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